

IUPAC name:

Prop-2-enamide

Molecular Formula:

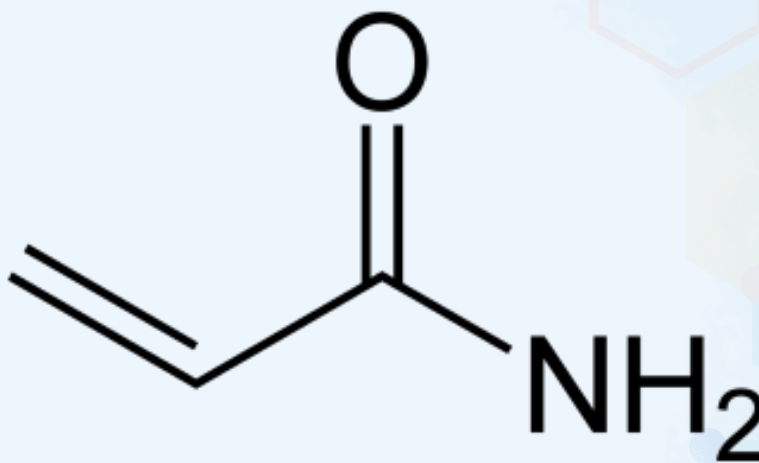
C₃H₅NO

Molar mass:

71.08 g/mol

Functional groups:

**alkene,
amide**



ACRYLAMIDE

"The Crispy Toxin"

Important Properties

- Bitter, odorless organic compound
- **Water-soluble;** highly soluble in water and polar organic solvents
- **Melting point:** 84.5 °C
- Contains polar amide groups that enable hydrogen bonding
- Toxic monomer used for polyacrylamide production

Uses/Value

- Used in wastewater treatment, papermaking, and oil recovery
- Added to chemical grouts and soil stabilizers
- Valuable industrial monomer for polyacrylamide production
- Provides synthesis material for lab polyacrylamide gels

Structure & Identity

- **Common name:** Acrylic Amide
- **Family:** Aliphatic amide
- **Appearance:** White crystalline solid
- **Natural sources:** Starchy foods cooked at high temperatures

Risks & Safety

- Classified as a probable human carcinogen (Group 2A) by the IARC
- Neurotoxic substance that can cause peripheral neuropathy in exposed workers
- The FDA recommends reducing dietary exposure by avoiding over-toasted or burnt foods
- Easily absorbed by the skin and respiratory tract, requiring strict lab safety PPE

What Is Amazing About It?

It plays a vital role in everyday paper manufacturing and advanced wastewater treatment systems. Its simple structure is synthesized directly from acrylonitrile and water.

Computer Engineering Relevance

Acrylamide synthesis relies on embedded automation systems to monitor critical thermal thresholds. Computer engineers design the real-time sensor networks and automated safety controls needed for its production.

References:

- International Agency for Research on Cancer. (1994). Some industrial chemicals. IARC Monographs on the Evaluation of the Carcinogenic Risk of Chemicals to Humans, 60. World Health Organization. <https://publications.iarc.who.int/Book-And-Report-Series/Iarc-Monographs-On-The-Identification-Of-Carcinogenic-Hazards-To-Humans/Some-Industrial-Chemicals-1994>
- Paluti, C., & Foist, L. (n.d.). Acrylamide definition, uses & structure. Study.com. <https://study.com/academy/lesson/acrylamide-definition-uses-structure.html>
- U.S. Food and Drug Administration. (2016, March). Guidance for industry: Acrylamide in foods. U.S. Department of Health and Human Services. <https://www.fda.gov/media/87150/download>
- Wikipedia. (2026, August 11). Acrylamide. <https://en.wikipedia.org/wiki/Acrylamide>